The Thales Project — Interim Review

2026-07-05 · paper trading only · a calibration ledger, not financial advice

Written the weekend before the first fills. Every order is still `accepted`, not filled — the market opens Monday 2026-07-06. This is therefore a review of a method and a book at the moment of truth, before the market has said anything back.

1. Executive summary

In two sessions we went from a set of API keys to a live paper-trading operation: a public GitHub repository, two custom tools, **eight three-legged positions** worth ~$40,210 of a $100,000 paper account, a written hypothesis for each, an event-based exit doctrine, a daily monitoring loop, and an audited skill that turns a raw hypothesis into a tradable structure.

The trading model is **not a close copy of any single named strategy**. Its fingerprint is a deliberate splice: the *structure* of regime-balanced investing[1], the *discipline* of scientific pre-registration[2], and the *objective* of a forecasting tournament[3] — where the goal is calibration[4], not profit. Where it most resembles a known financial strategy, the resemblance is partial and is stated as such in §4.

The report also does something the method demands of itself: it audits its own **integrity**. Our core loop is *hypothesis → test the market → take a viable position*. We drifted from it — one useful way, and three dangerous ones — and §6 names them all without flinching, because a calibration ledger that flatters itself is worthless. One of those three was caught only when an independent model reviewed this very report and found a defense in it that the repository disproves; that correction is left visible in §6.3 rather than quietly edited out, because how the dangerous drift *hides* is itself a finding.

2. How we got here

The path was incremental, and each step was a reaction to the last.

Connection and a single share. We stored the Alpaca[13] paper credentials, tested the connection, and bought one share of Apple — a plumbing test, never a thesis. It still sits in the ledger labelled "excluded from grading," which is itself a small act of the discipline to come: even the throwaway trade is logged honestly.

A ledger and a home. We built a repository around the Thales Playbook — the user's pre-existing method — and published it. The method was not invented here; it arrived in two PDFs. Our job was to *operationalise* it.

The playbook trios. Four trios came with the playbook already specified — Uranium, Electrons, Metals, Procurement. We enacted them at $5,000 each. Note the ordering, it matters later: **the positions came first; we wrote their hypotheses down afterward.**

The scout and the gauntlet. We scanned the tradable universe for new sectors, then ran six of them through the "gauntlet" — a pre-registered test that tries to *kill* each candidate before it can join the book. Four sectors yielded a surviving trio (Agriculture, Housing, GLP-1, Defense); two yielded *findings* — honest "no valid trio" verdicts (Space, AI-datacenter power). Crucially, here the ordering flipped: **the hypothesis and kill-criteria were written before the data was pulled.** This is the core model working as designed.

Exit doctrine and monitoring. We wrote event-based stop-losses (§ the exit-trigger report), had a second model[14] attack the alert rules for reliability, and set a daily loop to watch for the triggering events.

Tooling. Two skills now encode the method: one executes paper trades; one turns a hypothesis into a gauntlet-ready trio, with proxy fallbacks when a seat can't be filled directly. The second was tested and audited (11/11 vs 5/11 for a no-skill baseline) before this report.

The through-line: **start from a falsifiable idea, structure it, and let pre-declared criteria decide.** The moments we're proudest of and the moments we're wariest of both come from how faithfully we held that line.

3. What the model actually is

Every position is a **trio** — three assets filling three distinct *seats*:

- **Engine** — the asset that *is* the thesis. If the idea is right, this revenue line shows it.
- **Tollbooth** — collects regardless of which competitor wins (a fee, rent, royalty, or regulated return). Presses, not pickaxes[11].
- **Regime-inverse** — *feeds* when the thesis fails (not merely survives). This is the internal hedge: the leg designed to bleed in the base case and pay off in the bad one.

Weights are fixed at entry (42.5% / 32.5% / 25%), held for twelve months, no rebalancing. Before entry, each candidate must survive **five kill criteria**[5] — viability, authenticity, survival, self-funding, non-redundancy — and every kill must name the criterion that fired. Alongside the trios sits a small, hard-capped "cicada" sleeve[6] for rare dated-event bets; it is currently empty.

The stated product is **not returns**. It is calibration: a growing record of written predictions, pre-declared kill-lines, and honestly graded outcomes. "Green P&L on a dead mechanism is a rejection that got lucky" is the house motto, and it is the tell that this is a forecasting discipline wearing a portfolio's clothes.

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4. Which financial strategy is this most like?

Short answer: no single one — similarity is moderate at best, and the closest analogs are conceptual rather than mechanical. The table rates the resemblance honestly.

Known strategy	What it shares	Where it differs	Similarity
Permanent Portfolio[15] (Harry Browne)	The closest *structural* cousin: fixed weights, **no leverage**, no volatility-targeting, assets chosen so each *feeds* in a different macro regime, held passively	Browne balances regimes across the *whole portfolio* with four fixed sleeves; Thales does it *per theme per trio*, and adds a falsification test Browne has no equivalent of	**Moderate–high (on structure)**
All-Weather / risk parity[1]	The regime-inverse leg — holding assets that win in *different* macro regimes so the book isn't a single bet	Thales uses **no leverage** and no volatility-weighting[16], the defining machinery of risk parity; and balances within one theme, not across the book	**Moderate (concept only)**
Thematic / sector basket	Each trio is a concentrated bet on one sector's mechanism	A plain thematic basket has no internal hedge and no falsification test; Thales bolts both on	**Moderate**
"Picks and shovels" / value-chain investing[17]	The **tollbooth seat literally is this**: own the toll that collects whichever competitor wins, not the prospector	It's one seat of three, not the whole strategy; Thales pairs it with an engine and a hedge	**Moderate (on one seat)**
Long/short equity	The `short(N)` proxy rung, and the "inverse" language	Thales is a ~75/25 net-long *cross-hedge* within a theme, not a paired long/short book; shorts appear only as a last-resort synthetic seat; there is no market-neutral[18] target	**Low**
Barbell / tail-hedging[19]	The core-plus-cicada shape: a stable core plus tiny asymmetric bets	Thales's core is itself risk-bearing equity, not the "safe" end of a true barbell	**Low–moderate**
Event-driven / catalyst investing	Kill-lines and checkpoints are dated to real catalysts (earnings, legislation)	Event-driven funds trade the event; Thales uses events as *falsification dates*, not as the trade	**Low–moderate**
Superforecasting / forecasting tournaments[3]	The actual objective — calibration, pre-registration, honest grading, "grade the mechanism not the luck"	This is an *epistemology*, not a financial strategy; it says nothing about position construction	**High (on objective), N/A (on mechanics)**
Systematic/discretionary macro "thesis + stop"	Enter on a thesis, exit on a pre-defined invalidation	Macro desks size by conviction and trade continuously; Thales fixes weights and holds	**Moderate**

The honest synthesis. If forced to name it in one line: *a long-biased thematic equity book with an internal regime hedge, constructed and exited under science-style pre-registration, run to maximise forecasting calibration rather than return.* The three ingredients each exist in the wild; the **combination**, and especially the calibration objective, is what makes off-the-shelf labels fit poorly. Anyone who tells you this is "just risk parity" or "just thematic investing" is matching on one feature and ignoring the other two.

5. The book at the moment of truth (pre-trade charts)

These are built the same way as the gauntlet analyses — Alpaca monthly closes[7], dividend-adjusted, each trio treated as a fixed-weight book indexed to a starting value. They show the eight enacted trios at their **last pre-trade mark** — the July monthly bar through 2026-07-03, which (with Friday the observed Independence Day holiday) is effectively **Thursday's close**, before Monday's fills.

Two honesty notes before the numbers. First, **"YTD" here means "since the end-January monthly bar,"** because our monthly series indexes off the January *close* — it therefore omits January's own move, and is not the calendar-strict year-to-date a data terminal would show. Second, **the chart's own design invites the comparison the next section then disavows**: solid-vs-dashed lines and circle-vs-square markers visually sort gauntlet-born from playbook trios, tempting the eye to rank them — which §6.2(b) explains is exactly the artifact to distrust. The encoding is kept only to make the two *origins* legible, not to endorse the comparison.

Chart 1 — Year-to-date, indexed to January 2026

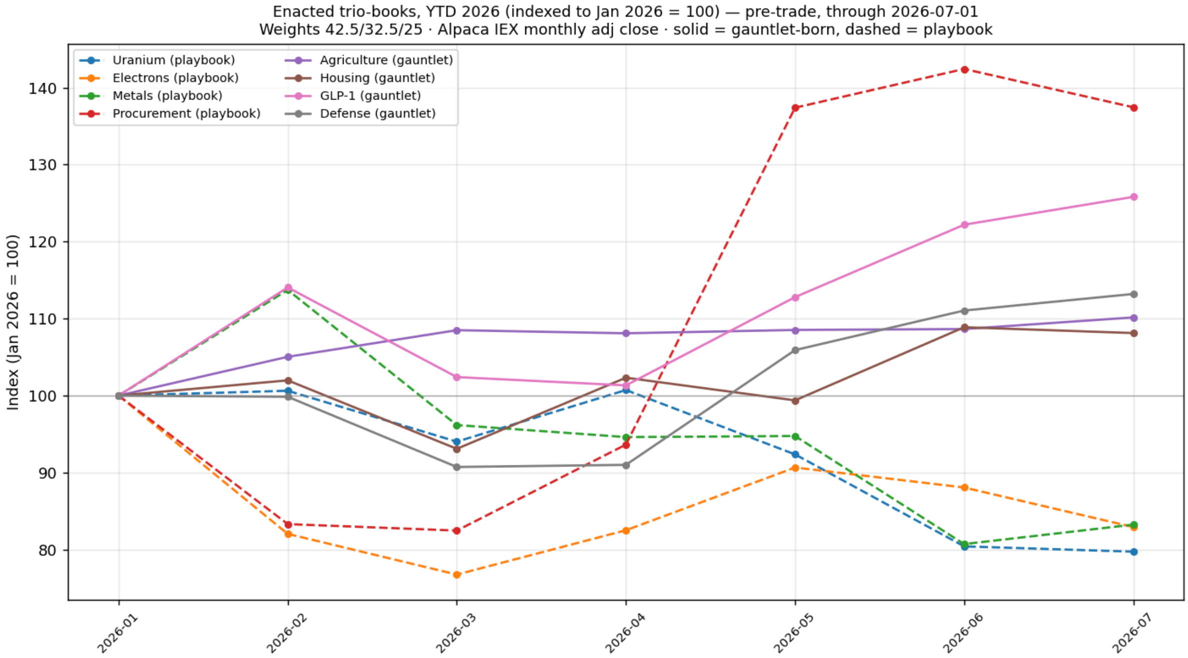
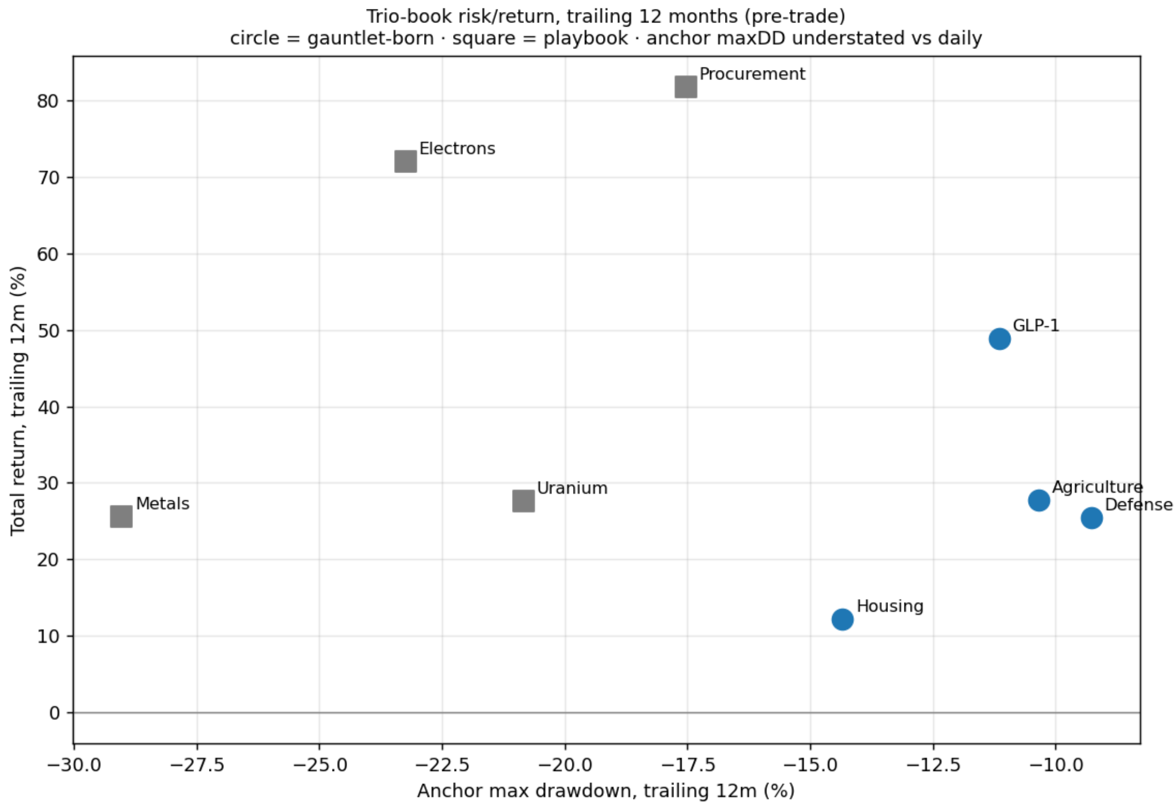


Chart 2 — Trailing-12-month return vs. maximum drawdown



The numbers behind the charts

Trio	Origin	YTD 2026	Trailing-12m return	Max drawdown[20]
Uranium	playbook	–20.3%	+27.7%	–20.8%
Electrons	playbook	–17.1%	+72.0%	–23.2%
Metals	playbook	–16.8%	+25.5%	–29.0%
Procurement	playbook	+37.4%	+81.8%	–17.5%
Agriculture	gauntlet	+10.2%	+27.7%	–10.3%
Housing	gauntlet	+8.1%	+12.2%	–14.3%
GLP-1	gauntlet	+25.8%	+48.9%	–11.2%
Defense	gauntlet	+13.2%	+25.4%	–9.3%

Read this table with suspicion, not satisfaction. The gauntlet-born trios (bottom four) look tidier — every one positive YTD, drawdowns clustered around –10%, while three of the four playbook trios are *negative* YTD with deeper drawdowns. It is tempting to conclude the gauntlet builds better trios. **That conclusion is largely circular, and §6 explains why.** (The Uranium row shows why single numbers mislead in both directions: –20.3% "YTD" but +27.7% over the trailing twelve months — not a typo, just two different windows, the sector having run hard through late 2025 and given some back since.) These charts are the pre-trade baseline against which the *forward* record — the only record that counts — will be graded from Monday onward.

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6. Integrity: the drift we watched for, and the drift that happened

Our core model is a straight line: **hypothesis → test the market → take a viable position.** Idea first; structure second; commitment last. We consciously tried not to drift from it. We drifted anyway, in two directions — and telling them apart is the whole point of this section.

6.1 The useful drift (upside): the gauntlet mass-produces trios

The gauntlet machinery let us take *six* sectors from raw idea to graded verdict in a single session, reusing one analysis engine, one data pipeline, one output shape. That is leverage of the good kind: more falsifiable structures per unit of effort, and — because two of the six honestly returned "no valid trio" — a method that produces **findings**, not just positions. Breadth like this is how a calibration ledger earns statistical power: ten dated predictions teach more than two. This drift we keep.

6.2 The dangerous drift (downside): reverse-engineering the hypothesis

Here is the uncomfortable part, stated plainly.

(a) We wrote some hypotheses *after* taking the positions. The four playbook trios were enacted first; their hypotheses (H1–H4 in the hypothesis register) were written down afterward, reverse-engineered from positions that already existed. In forecasting, this has a name — **HARKing**[8], Hypothesising After the Results are Known — and it is corrosive precisely *because it feels like insight*. A hypothesis fitted to a position you already hold can be phrased so it cannot fail; and a prediction that cannot fail produces **fake calibration**. You end up grading a forecast you never actually made in advance. For a project whose entire product is calibration, that is not a small blemish — it is the failure mode that voids the warranty.

(b) The charts in §5 select on the same window they're measured on. The gauntlet's survival test (K3) *kills* any candidate with a negative trailing-12-month return. So of course the gauntlet-born trios show positive trailing returns and shallow drawdowns — **they were chosen for exactly that.** Comparing them to the playbook trios, which were *not* filtered on Alpaca trailing data, and concluding the gauntlet is "better," is comparing a hand-picked sample to an unfiltered one on the very metric used to pick it. This is selection bias wearing the costume of a result. The tidy bottom four of the table are an artifact of the selection rule, not yet evidence of skill.

(c) The gauntlet doesn't just filter the window — it *searches* it (a third drift, caught in review). This report's first draft missed this; an independent review[9] found it. Two compounding problems:

- **The garden of forking paths.**[12] When a seat wouldn't fill, we *widened the pool and re-ran* — Housing's tollbooth was searched three times (BLDR → FNF, HD → ICE → ESNT) until one candidate survived the measurement window. Seat *contention* was also decided on trailing return and drawdown (ESNT beat MTG "on TR and DD"; ADM beat BG on drawdown; INVH beat AMH on TR). Drawing candidates until one passes the same window you then measure is multiple-comparisons selection — a stronger version of (b), not a separate mild one.
- **HARKing *inside* the gauntlet.** Housing's hypothesis (H6) is flagged in our own register as "the run's **discovered** hypothesis — the gauntlet found it by killing the flow-toll pool." A hypothesis formed from the data mid-run and enacted the same session is the same failure as §6.2(a), sitting in the half of the book we called clean.

6.3 What actually protects us — and one defense that doesn't

An earlier draft of this section claimed the gauntlet pre-registration files were "committed to version control **before** the data is pulled — a timestamped, tamper-evident record." **That claim was false, and the repository disproves it.** For all six gauntlets, `00-preregistration.md` and `01-run.md` landed in the *same git commit* (Agriculture in `1b19bba`, Housing in `20b2a12`, and so on). Git therefore provides **zero** tamper-evident separation between our hypotheses and our results. The pre-registration was real as a *workflow* — the prose was written before the data was pulled — but it is **unverifiable in the record**, and a single author controls his own commit timestamps anyway. Leaving the original claim in would have been precisely the self-flattery this ledger exists to refuse; catching it only in review is itself a small data point about how easily the dangerous drift hides.

So the honest list of what protects us is shorter than the draft pretended:

- The reverse-engineered hypotheses are marked as such — though bluntly: the register's *title* marks the **whole** thing "reverse-engineered from the book," with no per-hypothesis distinction, and the promised "discount for post-hoc origin" is so far a *promise, not a mechanism* — no discount rule is written down yet. That is a gap, not a safeguard.
- The one honest verdict on §5's charts is written into §5 itself: *read with suspicion.*

The **real** protection arrives Monday, and it has to carry the whole weight alone. Every position's grade date is **2026-08-15 and beyond** — genuinely out-of-sample[10]: after fills, on data that did not exist when the hypotheses were written. **The forward record cannot be reverse-engineered, cannot be searched, and cannot be back-dated.** The backward story is contaminated three ways over; the forward one is clean by construction. We grade on the forward one, and on nothing else.

Three standing rules adopted from this review, effective now, to keep the useful drift and kill the dangerous ones: (1) no position enters the book without a pre-registered, committed hypothesis *first* — the `hypothesis-to-trio` skill exists to enforce that ordering; (2) commit **and push** `00-preregistration.md` in its *own* commit before any data is pulled, so the next run's pre-registration is tamper-evident by construction rather than by assertion; (3) write an explicit grading discount for the post-hoc playbook hypotheses (H1–H4) before their first checkpoint, turning the promise into a mechanism. Integrity here is not politeness; it is the difference between measuring our judgement and flattering it.

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7. Current book and open threads

- **Deployed:** ~$40,210 of $100,000 (~40%), no leverage. Eight trios + one plumbing share, all `accepted`, filling at Monday's open.
- **All first checkpoints:** 2026-08-15. Nearest early trigger: Fastenal's mid-July monthly sales print.
- **Findings on the shelf:** four sectors gauntleted to "no valid trio" — Space and AI-datacenter power (from our own runs, reopen conditions logged), plus Water (inverse seat unfillable) and Shipping (killed on timing) inherited from the playbook.
- **Open threads:** (1) the daily monitoring loop currently lives in a session-scoped scheduler that expires in ~7 days — it needs promotion to a persistent routine; (2) Monday's fills must be written into the ledger with real entry prices to arm the price backstops; (3) the cicada sleeve is empty and awaits a qualifying dated event.

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8. Limitations

Backtests and the §5 charts rest on **monthly** closes from a **single, thinner data feed**[7], across **one regime of history** — a benign-to-strong year for most of these sectors. Drawdowns computed from monthly points are **understated** versus daily reality. Dividend-adjusted series flatter the raw-price experience of the yield-heavy names. None of the eight positions has filled yet, so there is **zero realised performance** to report — by design, this is a pre-trade document. And the whole enterprise is **paper**: no capital is at risk, which removes the single most important source of behavioural error in real trading, and that absence should temper any confidence drawn from what follows.

Reviewers are invited to attack §6 hardest of all — that is where the method lives or dies.

Footnotes

Glossary

- **Anchor** — a verified, dated price point used to reconstruct a return series when a full data feed isn't available.
- **Authenticity (K2)** — a kill criterion: the asset must genuinely embody its seat, not merely be adjacent (e.g. a conglomerate that happens to touch the theme fails).
- **Backtest** — simulating how a fixed set of positions would have performed over past data. Evidence, never proof; Thales treats a winning backtest as killable.
- **Book** — the whole portfolio, or a single trio treated as a mini-portfolio with fixed weights.
- **Calibration** — see footnote 4. The project's true objective.
- **Cicada sleeve** — see footnote 9.

- **Drawdown** — a fall from a prior peak; "max drawdown" is the worst such fall in a window.
- **Engine** — the trio seat that *is* the thesis; carries the most weight (42.5%).
- **Gauntlet** — the pre-registered process of trying to kill each candidate before it can join the book; survivors become positions, and a sector that fills no valid trio becomes a logged *finding*.
- **HARKing** — see footnote 13. The integrity failure of writing hypotheses after seeing results.
- **Kill criteria (K1–K5)** — see footnote 8. The five pre-declared tests every candidate must survive.
- **Kill-line** — a pre-declared, specific event that, if observed, closes a position; the exit-doctrine equivalent of a stop-loss, but triggered by *evidence*, not price.
- **Market-neutral** — see footnote 20. A book with market moves balanced out; *not* a Thales target.
- **Notional** — the dollar size of a position (here, e.g., $2,125 for an engine leg), independent of share count.
- **Out-of-sample** — see footnote 16. Data not used to build or pick the model; the only honest test of a forecast.
- **P&L** — "profit and loss," the running gain or loss on a position. "Green P&L" simply means it's up.
- **Paper trading** — simulated trading with fake money against real prices; no capital at risk.
- **Playbook trio** — one of the four trios inherited pre-specified from the Thales Playbook PDFs (Uranium, Electrons, Metals, Procurement), enacted before their hypotheses were formalised.
- **Pre-registration** — see footnote 2.
- **Proxy ladder** — the fallback sequence when a seat can't be filled directly: a listed proxy (e.g. an ETF), then a synthetic short (`short(N)`), then declaring the seat unfillable.
- **Regime** — a broad macro-economic state (e.g. high inflation, or an industrial recovery). The regime-inverse seat is defined by *which* regime helps it.
- **Regime-inverse** — the trio seat that *feeds* when the thesis fails; the internal hedge. Must feed, not merely dodge.
- **`short(N)`** — notation for a synthetic seat built by short-selling asset N (profiting if N falls); used only as a last-resort proxy, with its costs (it hedges rather than feeds; carries borrow cost) logged.
- **Tollbooth** — the trio seat that collects regardless of which competitor wins; prefers a toll on the installed *stock* over a toll on transaction *flow*.
- **Trio** — the three-seat unit (engine + tollbooth + regime-inverse) that is the project's atomic position.
- **Uninsured quadrant** — the one scenario that hurts all three seats of a trio at once; named in advance for every position so the blind spot is known, not discovered.

All-Weather / risk parity — an approach (associated with Bridgewater's All-Weather fund) that holds assets balanced to perform across different economic "regimes" (growth/inflation up or down), typically using leverage to equalise each asset's risk contribution. Thales borrows the *regime-balancing idea* but not the leverage or the risk-equalising math.

Pre-registration — the scientific practice of publicly recording a hypothesis and analysis plan *before* collecting data, so results can't be retrofitted to a story. Borrowed here via committing kill-criteria to git before pulling prices.

Forecasting tournament / superforecasting — structured prediction contests (Philip Tetlock's research) where success is measured by *calibration and accuracy of stated probabilities*, not profit. Thales's objective function is this, not a fund's.

Calibration — the degree to which stated confidence matches reality: if you say 70% and you're right 70% of the time, you're well-calibrated. The project's actual deliverable.

Five kill criteria (K1–K5) — viability (real size, not a bankruptcy candidate), authenticity (the asset genuinely *is* the thesis seat), survival (positive trailing year, plus a max-drawdown floor), self-funding (no forced dilutive capital raise), and non-redundancy (not too correlated with another seat). Note a real inconsistency in our own corpus: the *gauntlet* runs apply the survival floor at ~–60% drawdown, while the *playbook* entries state it as "round-tripped ~–75%." Both appear in the repository; they should be reconciled to one number in the next between-runs amendment. See the glossary for each criterion.

Cicada sleeve — a small (<5% of book), hard-capped allocation to rare events whose date is fixed in advance (like a periodical cicada emergence), which the market has forgotten to price. Currently empty.

Data feed (IEX vs SIP) — Alpaca's free tier serves prices from the IEX exchange only (thinner volume) rather than the consolidated SIP tape (all exchanges), so monthly closes can differ marginally from the "official" consolidated close.

HARKing — "Hypothesising After the Results are Known." Presenting a hypothesis invented to fit already-seen data as if it had been predicted in advance. A recognised research-integrity failure; the central risk §6 guards against.

Independent review — this report was reviewed by a separate Claude model (Fable) before publication; §6.2(c) and the correction in §6.3 are its findings, adopted. ■

Out-of-sample — evaluated on data that was *not* used to build or select the model. The opposite of testing on the same data you fitted to; the only honest test of a prediction. Monday's fills onward are out-of-sample by construction. ■

"Presses, not pickaxes" — the Thales metaphor: in a gold rush, sell the tools (or here, own the toll) rather than dig. Thales of Miletus cornered the olive *presses*. ■

Garden of forking paths — a research-integrity term (Andrew Gelman) for the bias introduced when the analysis path is chosen *after* seeing the data — including trying alternative candidates until one "works." Distinct from outright cherry-picking because each individual choice can feel principled. ■

Alpaca — a brokerage with an API for automated trading; its *paper* environment simulates fills with fake money against real market data. ■

A second model — the alert rules were independently inspected by a different Claude model (Fable) to reduce single-author blind spots. ■

Permanent Portfolio — Harry Browne's fixed four-way split (stocks / long bonds / gold / cash), unleveraged and rebalanced to fixed weights, each sleeve designed to carry a different macro regime. Structurally the closest passive cousin to a trio. ■

Volatility-weighting — sizing positions so each contributes equal *risk* (volatile assets get smaller weights). Core to risk parity; absent in Thales, which uses fixed 42.5/32.5/25 weights regardless of volatility. ■

"Picks and shovels" / value-chain investing — buying the suppliers/enablers of a boom rather than its direct players (sell shovels to prospectors). The tollbooth seat is this idea in one leg. ■

Market-neutral — a book balanced so overall market moves wash out, leaving only the relative bet. Thales does *not* target this; it stays net-long. ■

Barbell / tail-hedging — an allocation (associated with Nassim Taleb) that combines very safe assets with a few high-payoff, low-cost bets, avoiding the "medium risk" middle. ■

Maximum drawdown — the largest peak-to-trough drop over the window, expressed as a percentage. Here computed from monthly points, so it *understates* the intra-month lows a daily series would show. ■